Looking Upon the Seven Hills and Beyond

Insights from San Francisco's Property Tax Roll Data — Through the Lens of AI

There's something uniquely compelling about looking at a city from above. You begin to see patterns and relationships that aren't obvious at street level. In this analysis, that vantage point isn't physical—**it's data**.

San Francisco has accumulated a vast amount of detailed public data over time. The challenge today isn't access—it's interpretation. **How do we turn what's already available into insights that actually matter?**

Traditionally, that requires a combination of statistics, programming, and domain expertise. However, even experienced professionals often run into a practical constraint—**coding takes time**. Testing ideas, especially on a new dataset, can be slow and iterative. You may have a strong intuition or a promising angle, but validating it could require hours (or even days) of work. As a result, **many good ideas simply don't get explored**.

AI changes that dynamic.

With the right prompts, AI tools like ChatGPT can generate working analysis code quickly. The focus shifts away from writing code and toward asking better questions and interpreting results. That doesn't diminish the role of human judgment—if anything, it makes it more important. AI handles the execution; we stay responsible for direction and interpretation. It's a partnership: **AI extends our reach, but we decide where to look and what matters**.

This project is a practical example of that approach.

With a background in data science and a recent move to San Francisco, I set out to explore the city through its property tax roll data, available via **DataSF**. I started with a rough analytical plan—key questions I wanted to answer, and hypotheses worth testing. From there, **ChatGPT** was used to generate an initial version of the analysis code.

That code was tested, adjusted, and improved. As the first set of visualizations came together, they led to new questions—some expected, others less so. Each iteration refined the analysis further. From a cold start, the first meaningful results were produced in under an hour, and the work evolved from there into a more complete and polished report.

What follows is a summary of the key findings. I'll briefly outline the analytical approach and reasoning, but the emphasis is on results—what the data shows, and what we can take away from it. The full code is available on GitHub for those interested.

GitHub Repository:
`https://github.com/msmsimon00/sf-assessor-2-circuits`

Let's get into it.

San Francisco Assessor Roll Analysis

A portfolio-style structural study of value, neighborhood geography, and the Two Housing Circuits of San Francisco housing

source: https://thefrontsteps.com/san-francisco-real-estate-district-maps/

Introduction

This project turns the San Francisco historical secured property tax roll into a geospatial housing analysis.

The objective is not to estimate exact market prices, but to use the roll to explain how the city's housing stock is distributed, how assessments differ across neighborhoods, and which parts of the stock actually circulate through the market.

The central conclusion is straightforward:

> **San Francisco does not behave like one unified housing market. It behaves like two overlapping housing circuits:**
>
> - a **legacy circuit** of older, tightly held, slower-moving homes; and
> - a **Circulation Circuit** of homes that return to the market more frequently and absorb a disproportionate share of new demand.

The analysis builds that argument step by step using maps, value comparisons, building age, recent-sales filters, turnover measures, owner-occupancy proxies, and holding-period distributions.

Project motivation

The San Francisco secured property tax roll is a large administrative dataset. On its face, it is about assessed values, parcel identifiers, and land-use categories. But with careful cleaning and interpretation, it can do much more:

- turn a raw roll into a usable geospatial housing dataset;
- show how San Francisco's property stock is distributed across neighborhoods and property types;
- distinguish between the **full stock of housing** and the **subset of housing that actually circulates through the market**;
- reveal why two homes in the same city can belong to very different economic worlds.

This notebook approaches the assessor roll as a city-structure dataset rather than just a valuation table. The central objective is to understand not only **how much housing exists**, but **which parts of that housing stock are active, liquid, owner-oriented, investor-oriented, old, new, stable, or recycling through turnover**.

What this analysis does

The notebook proceeds in a deliberate sequence so that the later interpretations rest on transparent data choices rather than black-box modeling.

1. It loads the historical secured roll using an explicit schema so fields keep the intended types.
2. It parses geometry and converts the assessor roll into a geospatial dataset.
3. It tests whether small auxiliary value fields such as fixtures and personal property are material enough to matter for broad analysis.
4. It defines a core **total assessed value** measure based on land value plus improvement value.
5. It summarizes San Francisco's property stock by broad property category.
6. It maps where those categories sit in the city and examines how their counts evolve through time.
7. It compares property categories not only by parcel counts but also by **median parcel value** and **aggregate tax-base contribution**.
8. It narrows the focus to **single-family homes** and **condominiums**, the two housing segments most relevant to owner entry, market competition, and neighborhood comparison.
9. It studies those two housing segments through five complementary lenses:
   - total assessed value,
   - assessed value per square foot,
   - building age,
   - recent transactions,
   - owner-occupied versus renter/investor-oriented behavior.
10. It ends with a dynamic turnover analysis based on **years since last sale** and exponential decay fitting.

Why the methodology matters

This project is not trying to estimate true market prices from assessor data. It is doing something more structural:

- **Assessed value** is used as a citywide comparative signal, especially helpful for neighborhood ranking and tax-base structure.
- **Recent-sales filtering** is used because under **Proposition 13**, long-held properties may have assessments anchored to old base-year values.
- The **homeowner exemption** is used as an **owner-occupied proxy**, not as a perfect legal classification.
- **Years since last sale** is used to distinguish housing that is present in the city from housing that is actually participating in the city's current exchange process.

That distinction becomes crucial later. A city can have a large housing stock on paper while still offering only a thin stream of homes for actual purchase. This report shows why.

Important interpretation note

California's assessment structure matters throughout this report. Under Proposition 13, properties that have not changed hands in many years can carry assessed values far below current market value. By contrast, recently purchased homes are usually reassessed closer to contemporary market conditions. That means:

- long-held housing stock tells a story about **continuity and embedded ownership history**;
- recently sold stock tells a story that is **closer to the current market**;
- the gap between the two is itself economically meaningful.

The report therefore separates **headline housing stock** from **circulating housing stock** whenever possible.

Section 1. Citywide property structure

The assessor roll contains many detailed use codes. For citywide structure, the notebook groups them into broader categories such as:

- single-family residential,
- multi-family residential,
- retail,
- office,
- hotel,
- industrial,
- public / vacant-government land.

The first question is simple but important: what kinds of properties make up San Francisco's built environment, and where are they located?

Key findings

The neighborhood concentration pattern is strongly consistent with San Francisco's built form:

- **Single-family homes** are concentrated in outer and hill-oriented neighborhoods such as Bernal Heights, Parkside, Central Sunset, and Excelsior.
- **Multi-family housing** is densest in urban neighborhoods such as Inner Mission, Central Richmond, Noe Valley, Inner Richmond, and Inner Sunset.
- **Office and hotel uses** cluster in the downtown core—especially Financial District North, Union Square, Financial District South, South of Market, and Civic Center.
- **Industrial parcels** concentrate in Bayview, Potrero Hill, South of Market, and Dogpatch / Central Waterfront.
- **Government** / **vacant land** has its own distinct geography and does not behave like ordinary taxable housing stock.

The map matters because raw counts alone cannot show whether a category is corridor-based, waterfront-oriented, diffuse, or core-concentrated.

Figure 1. Citywide property-type map

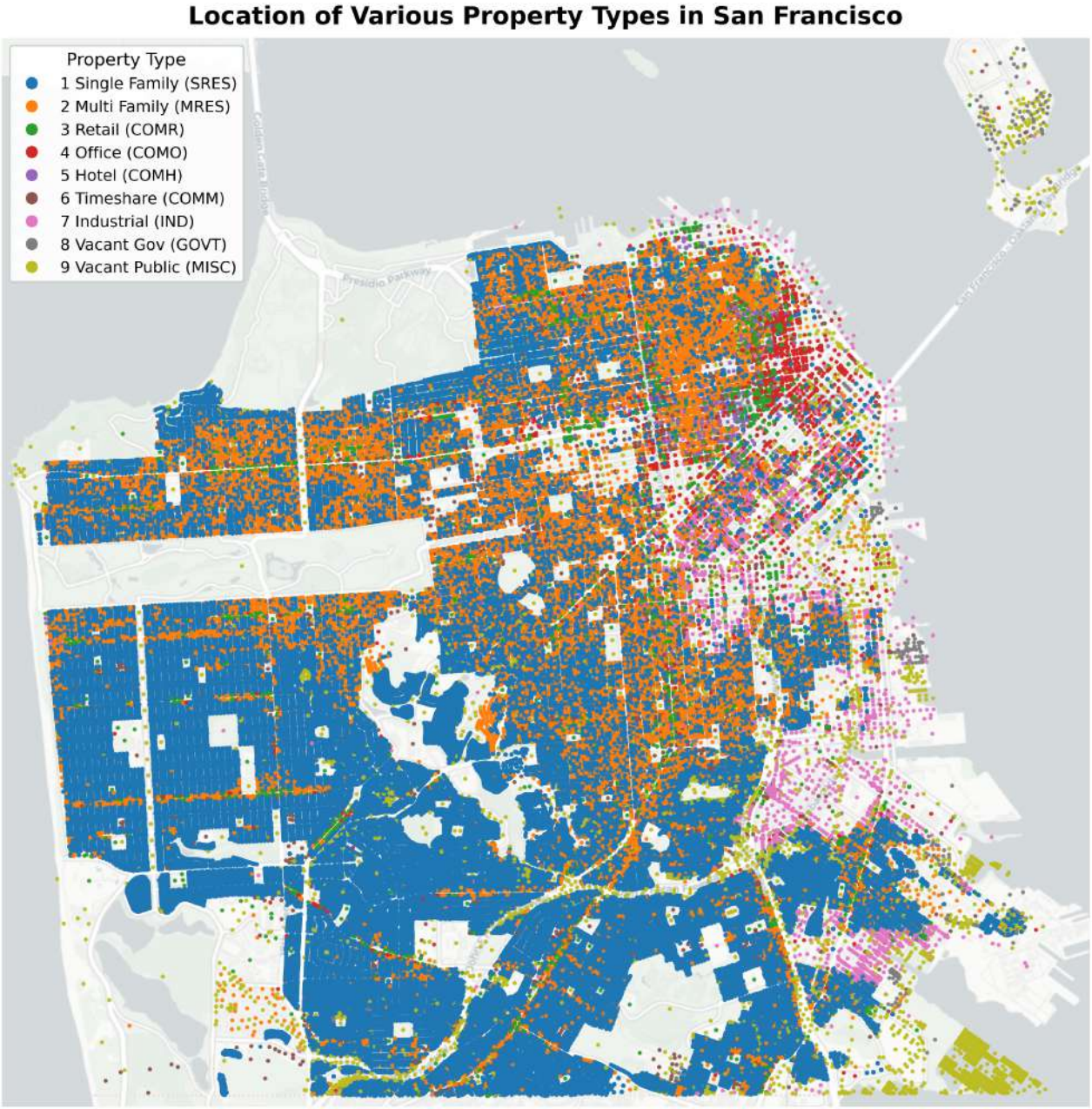


Table 1. Top 5 neighborhoods by property count within each property type

1 Single Family (SRES)	2 Multi Family (MRES)	3 Retail (COMR)	4 Office (COMO)	5 Hotel (COMH)
9A Bernal Heights	9C Inner Mission	9C Inner Mission	8B Financial District North	8B Financial District North
2D Parkside	1A Central Richmond	8B Financial District North	8I Union Square	8J Tenderloin
2E Central Sunset	5C Noe Valley	9F South of Market	9F South of Market	9F South of Market
9H South Beach	1B Inner Richmond	8I Union Square	9B Financial District South	8A Downtown
10C Excelsior	2F Inner Sunset	6C Lower Pacific Heights	8F Van Ness/ Civic Center	8I Union Square

6 Timeshare (COMM)	7 Industrial (IND)	8 Vacant Gov (GOVT)	9 Vacant Public (MISC)
8A Downtown	10A Bayview	9I Treasure Isl./Yerba Buena Isl.	10M Candlestick Point
6B Hayes Valley	9F South of Market	10J Hunters Point	10A Bayview
8E Russian Hill	9C Inner Mission	10A Bayview	9A Bernal Heights
9C Inner Mission	9E Potrero Hill	9J Central Waterfront/Dogpatch	10J Hunters Point
8F Van Ness/ Civic Center	9J Central Waterfront/Dogpatch	9D Mission Bay	10D Outer Mission

Section 2. How property types change over time

This section shifts from a single-year map to long-run structure.

Key findings

The roll composition is surprisingly stable, but not static.

- In **2024**, the roll contains **154,886 single-family** records and **36,320 multi-family** records, confirming that residential parcels dominate the dataset by count.
- Single-family records fell from **140,514 in 2010** to **132,843 in 2011**, a drop of about **5.5%**, before recovering and reaching a new high by 2024.
- Multi-family records drifted down modestly from **37,941 in 2007** to **36,320 in 2024**, a decline of roughly **4.3%**.
- Timeshare properties decline materially after 2020, which is plausibly related to pandemic-era stress in tourism-linked uses.
- Smaller specialist categories are more cyclical. A few hundred records can look dramatic in percentage terms even when they remain small relative to the citywide stock.

The broad takeaway is that **San Francisco's property inventory changes slowly**. That matters because many contemporary housing debates are driven not by rapid stock replacement, but by pressures imposed on a stock that is already largely built out.

Figure 2. Property counts over time by category

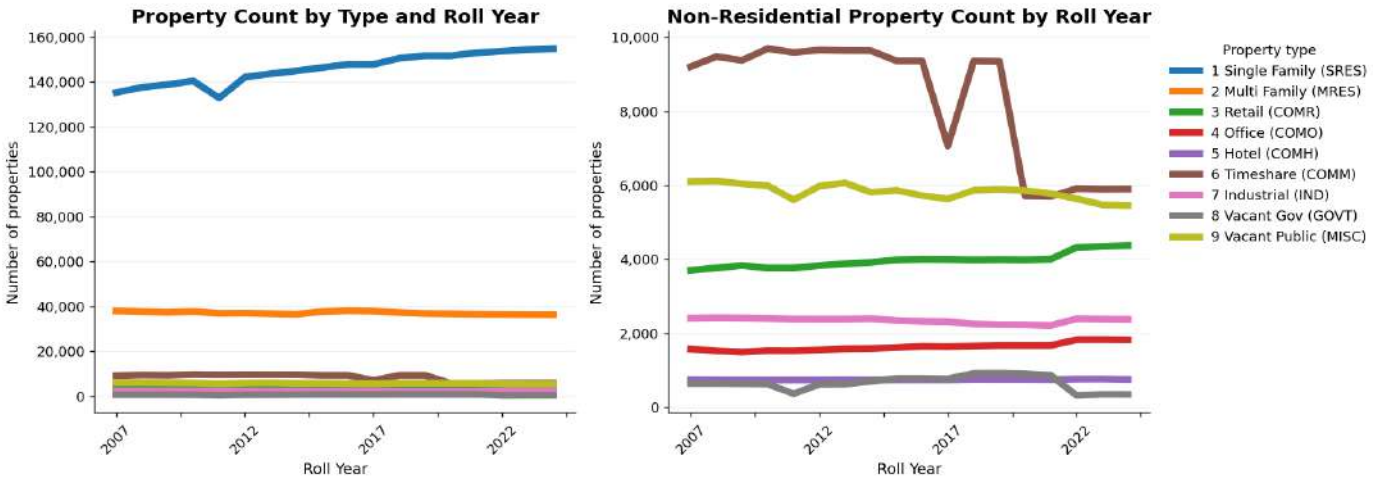
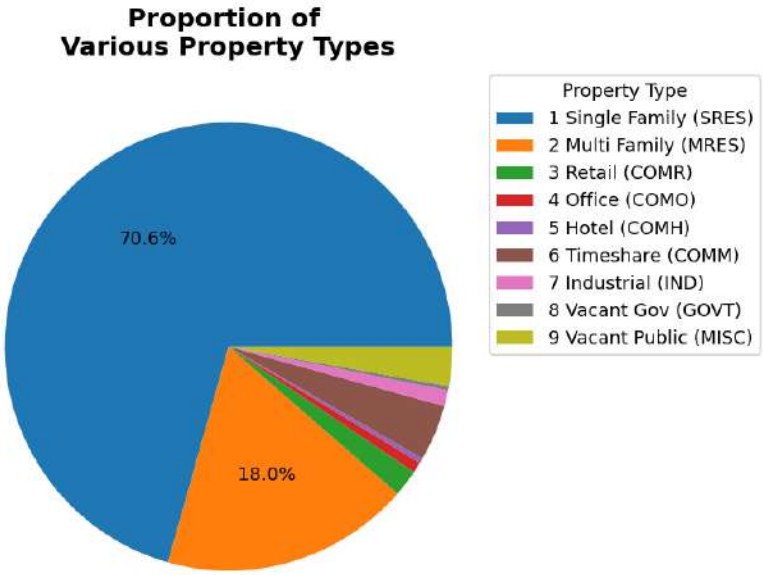


Figure 3. Average property-type shares over time



Section 3. Assessed values by property type

Parcel counts alone cannot show economic importance. A category may matter because it is:

- extremely numerous,
- expensive per parcel,
- or both.

This section therefore separates two questions:

1. **What is the median assessed value of a typical parcel in each category?**
2. **What is the aggregate assessed value contribution of that category to the city's tax base?**

Key findings

At the individual-parcel level:

- The median assessed value of a **single-family** property rises from **$0.316M in 2007** to **$0.747M in 2024**, an increase of about **136%**.
- The median **multi-family** property rises from **$0.426M** to **$0.994M**, up roughly **133%**.
- Office and hotel parcels remain expensive on a per-record basis, reaching roughly **$2.2M per office parcel** and **$3.0M per hotel parcel** in 2024.
- Government-owned vacant parcels often carry very low or zero assessed values, consistent with exemption or distinct administrative treatment.

At the aggregate citywide level:

- Total assessed value for **single-family housing** increases from **$58.1B in 2007** to **$154.0B in 2024**.
- **Multi-family housing** increases from **$25.8B** to **$74.7B**.
- **Office** begins from a very large **$18.4B** base in 2007 and rises to **$59.7B**, despite far fewer parcels than the residential categories.

Figure 4. Median and total assessed value by property type

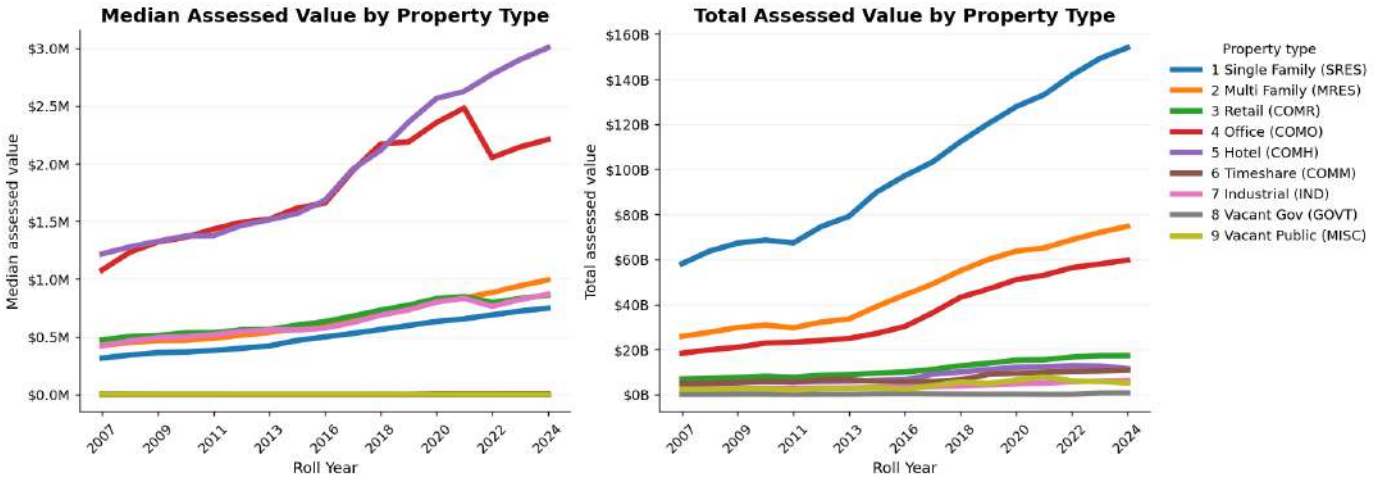


Table 2. Median assessed value by property type and roll year

Roll Year	1 Single Family (SRES)	2 Multi Family (MRES)	3 Retail (COMR)	4 Office (COMO)	5 Hotel (COMH)	6 Timeshare (COMM)	7 Industrial (IND)	8 Vacant Gov (GOVT)	9 Vacant Public (MISC)
2007	$0.316M	$0.426M	$0.474M	$1.079M	$1.218M	$0.006M	$0.421M	$0.000M	$0.001M
2008	$0.342M	$0.449M	$0.502M	$1.233M	$1.279M	$0.005M	$0.464M	$0.000M	$0.001M
2009	$0.363M	$0.469M	$0.511M	$1.322M	$1.327M	$0.004M	$0.492M	$0.000M	$0.001M
2010	$0.368M	$0.470M	$0.534M	$1.361M	$1.373M	$0.004M	$0.499M	$0.000M	$0.001M
2011	$0.382M	$0.487M	$0.535M	$1.433M	$1.376M	$0.004M	$0.519M	$0.000M	$0.001M
2012	$0.400M	$0.513M	$0.560M	$1.492M	$1.464M	$0.003M	$0.546M	$0.000M	$0.001M
2013	$0.421M	$0.538M	$0.563M	$1.517M	$1.513M	$0.003M	$0.560M	$0.000M	$0.001M
2015	$0.468M	$0.570M	$0.600M	$1.614M	$1.567M	$0.003M	$0.558M	$0.000M	$0.001M
2016	$0.499M	$0.589M	$0.631M	$1.657M	$1.682M	$0.003M	$0.575M	$0.000M	$0.001M
2017	$0.529M	$0.651M	$0.679M	$1.941M	$1.954M	$0.003M	$0.623M	$0.000M	$0.001M
2018	$0.563M	$0.698M	$0.730M	$2.170M	$2.115M	$0.004M	$0.686M	$0.000M	$0.001M
2019	$0.597M	$0.757M	$0.773M	$2.188M	$2.359M	$0.003M	$0.732M	$0.000M	$0.001M
2020	$0.632M	$0.803M	$0.830M	$2.355M	$2.564M	$0.008M	$0.800M	$0.000M	$0.001M
2021	$0.655M	$0.835M	$0.847M	$2.481M	$2.626M	$0.007M	$0.832M	$0.000M	$0.000M
2022	$0.689M	$0.883M	$0.796M	$2.053M	$2.775M	$0.007M	$0.763M	$0.000M	$0.000M
2023	$0.722M	$0.942M	$0.830M	$2.144M	$2.901M	$0.008M	$0.821M	$0.000M	$0.000M
2024	$0.747M	$0.994M	$0.859M	$2.209M	$3.005M	$0.008M	$0.867M	$0.000M	$0.000M

Section 4. Focus on single-family homes and condominiums

From this point onward, the notebook narrows the lens to the two housing segments that matter most for entry, neighborhood comparison, and market turnover:

- **Single-family homes**
- **Condominiums**

This narrower focus is analytically useful because these are the property types households actually try to buy. The report also filters out implausibly small land, improvement, and total assessments so that vacant land, incomplete records, and non-comparable outliers do not distort comparisons.

4A. Where are the expensive and affordable neighborhoods?

Key findings

For **single-family homes**:

- **Presidio Heights** leads at **$4.729M**.
- **Hunters Point** sits at **$0.342M**.
- That is a spread of roughly **13.8x** within the same city.

For **condominiums**:

- **Presidio Heights** still leads, at **$1.605M**.
- The low end is **Mount Davidson Manor** at **$0.449M**.
- The spread is much narrower, about **3.6x**.

This difference is revealing. Single-family values combine:neighborhood prestige, lot size, structure size, and embedded scarcity. Condo values are still highly unequal, but their narrower spread suggests a stronger role for: location, building typology,and urban access rather than lot scale.

Figure 5. 2024 assessed value maps: single-family vs condominiums

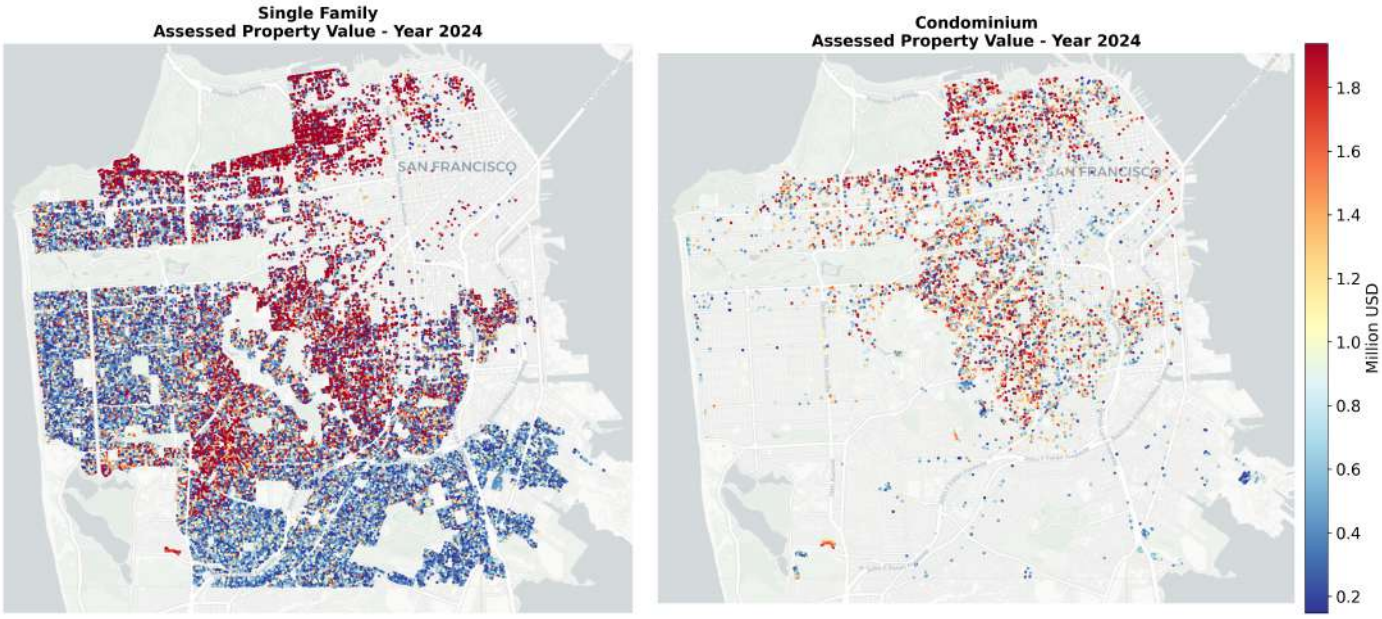


Table 3: Single-family neighborhoods with highest median assessed value (Year 2024)

Neighborhood	USD millions	Property count
7C Presidio Heights	$4.729M	553
7B Pacific Heights	$4.160M	1032
7D Cow Hollow	$2.542M	474
1F Sea Cliff	$2.433M	466
7A Marina	$2.274M	638
8E Russian Hill	$2.062M	322
5H Clarendon Heights	$2.049M	288
8G Telegraph Hill	$1.972M	106
5F Buena Vista Park/Ashbury Heights	$1.896M	442
6C Lower Pacific Heights	$1.786M	345

Table 4: Single-family neighborhoods with lowest median assessed value (Year 2024)

Neighborhood	USD millions	Property count
10J Hunters Point	$0.342M	146
10E Visitacion Valley	$0.411M	3250
10K Bayview Heights	$0.435M	551
10G Silver Terrace	$0.442M	1935
10A Bayview	$0.445M	2732
10C Excelsior	$0.450M	4643
3H Ingleside	$0.453M	1815
10N Little Hollywood	$0.455M	399
10B Croker Amazon	$0.464M	2521
10F Portola	$0.478M	3814

Table 5: Condominium neighborhoods with highest median assessed value (Year 2024)

Neighborhood	USD millions	Property count
7C Presidio Heights	$1.605M	339
3D Stonestown	$1.420M	122
7D Cow Hollow	$1.200M	1056
9B Financial District South	$1.174M	2438
1C Jordan Park/Laurel Heights	$1.167M	239
7A Marina	$1.138M	1111
7B Pacific Heights	$1.136M	2786
9H South Beach	$1.127M	3671
8E Russian Hill	$1.098M	1617
5G Corona Heights	$1.093M	533

Table 6: Condominium neighborhoods with lowest median assessed value (Year 2024)

Neighborhood	USD millions	Property count
4N Mount Davidson Manor	$0.449M	129
10B Croker Amazon	$0.467M	242
8J Tenderloin	$0.522M	406
4B Diamond Heights	$0.547M	668
3G Ingleside Heights	$0.587M	368
8A Downtown	$0.594M	1201
10A Bayview	$0.601M	215
10J Hunters Point	$0.609M	581
8H North Waterfront	$0.629M	559
2C Outer Sunset	$0.645M	117

4B. What about value per square foot?

Total assessed value is partly a size story. To separate size from value intensity, the notebook turns to **assessed value per square foot**.
This is not the same as observed market sale price per square foot, but it is highly informative because it distinguishes:

- neighborhoods where homes are valuable mainly because they are **large**, and
- neighborhoods where each unit of housing space is itself **highly valued**.

Key findings

For **single-family homes**:

- The most expensive space appears in **Telegraph Hill ($1,199/sq ft)**, **Pacific Heights ($1,167/sq ft)**, and **Presidio Heights ($1,134/sq ft)**.
- The low end is concentrated in the southeast: **Hunters Point ($253/sq ft)**, **Bayview ($305/sq ft)**, and **Visitacion Valley ($322/sq ft)**.

For **condominiums**:

- The top tier is distinctly downtown and redevelopment oriented: **Financial District South ($1,145/sq ft)**, **Central Waterfront / Dogpatch ($1,145/sq ft)**, **South Beach ($1,142/sq ft)**, and **Mission Bay ($1,047/sq ft)**.

This is a strong result:

- The condo market appears to price access to **employment, waterfront redevelopment,** and **dense amenity clusters** more directly than the single-family market does.
- The single-family market still prices prestige neighborhoods very highly, but its geography remains less exclusively downtown-oriented.

Figure 6. Assessed value per square foot maps

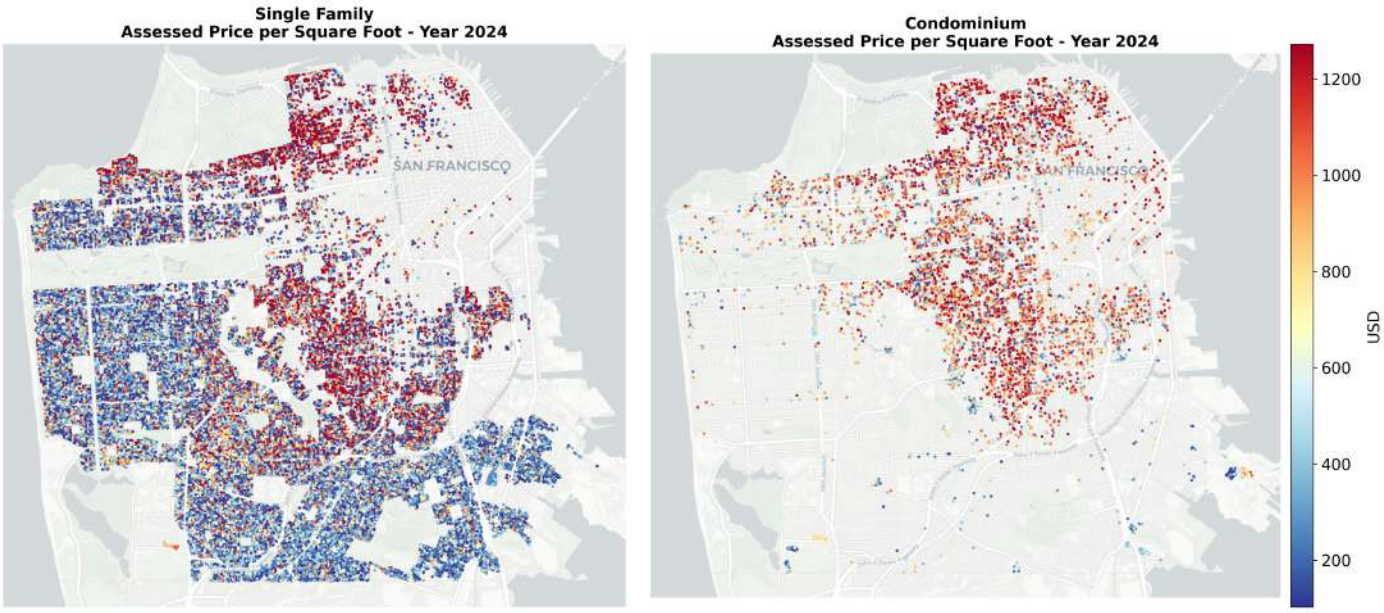


Table 7: Single-family neighborhoods with highest assessed value per square foot (Year 2024)

Neighborhood	Value per sq ft	Property count
8G Telegraph Hill	$1,199 / sq ft	106
7B Pacific Heights	$1,167 / sq ft	1032
7C Presidio Heights	$1,134 / sq ft	553
7D Cow Hollow	$1,089 / sq ft	474
8E Russian Hill	$1,060 / sq ft	322
6C Lower Pacific Heights	$932 / sq ft	345
7A Marina	$924 / sq ft	638
1F Sea Cliff	$864 / sq ft	466
5K Eureka Valley/Dolores Heights	$862 / sq ft	1498
5F Buena Vista Park/Ashbury Heights	$841 / sq ft	442

Table 8: Single-family neighborhoods with lowest assessed value per square foot (Year 2024)

Neighborhood	Value per sq ft	Property count
10J Hunters Point	$253 / sq ft	146
10A Bayview	$305 / sq ft	2732
10N Little Hollywood	$317 / sq ft	399
10K Bayview Heights	$319 / sq ft	551
10E Visitacion Valley	$322 / sq ft	3250
10B Croker Amazon	$329 / sq ft	2521
10G Silver Terrace	$333 / sq ft	1935
4B Diamond Heights	$334 / sq ft	475
1A Central Richmond	$336 / sq ft	2228
10F Portola	$343 / sq ft	3814

Table 9: Condominium neighborhoods with highest assessed value per square foot (Year 2024)

Neighborhood	Value per sq ft	Property count
9B Financial District South	$1,145 / sq ft	2438
9J Central Waterfront/Dogpatch	$1,145 / sq ft	715
9H South Beach	$1,142 / sq ft	3671
7A Marina	$1,096 / sq ft	1111
7D Cow Hollow	$1,092 / sq ft	1056
9K Yerba Buena	$1,088 / sq ft	836
9D Mission Bay	$1,047 / sq ft	2223
8E Russian Hill	$1,034 / sq ft	1617
7C Presidio Heights	$1,033 / sq ft	339
6B Hayes Valley	$1,028 / sq ft	1839

Table 10: Condominium neighborhoods with lowest assessed value per square foot (Year 2024)

Neighborhood	Value per sq ft	Property count
4N Mount Davidson Manor	$355 / sq ft	129
10B Croker Amazon	$440 / sq ft	242
3A Lake Shore	$451 / sq ft	200
2C Outer Sunset	$559 / sq ft	117
10A Bayview	$569 / sq ft	215
2E Central Sunset	$580 / sq ft	158
1E Outer Richmond	$582 / sq ft	517
4B Diamond Heights	$649 / sq ft	668
10M Candlestick Point	$656 / sq ft	375
10J Hunters Point	$661 / sq ft	581

4C. How old are the buildings?

Building age adds a structural time dimension. It can proxy for:

- development era,
- redevelopment pressure,
- preservation,
- prestige,
- functional obsolescence,
- and the degree to which housing stock is a legacy inheritance versus a newer supply wave.

Key findings

For **single-family housing**:

- The oldest neighborhoods in the table—**Lower Pacific Heights, Hayes Valley, North Panhandle, Inner Mission, and Haight Ashbury**—cluster around a median age of **124 years**.
- The youngest single-family neighborhood in the filtered table is **Hunters Point** at **36 years**, followed by **Diamond Heights (59 years)** and **Forest Knolls (63 years)**.

For **condominiums**:

- The youngest stock is concentrated in redevelopment districts:
  Hunters Point (8 years) Stonestown (9 years) Central Waterfront / Dogpatch (12 years) Financial District South (15 years) Mission Bay (15 years)

This contrast is one of the report's most durable findings:

> **The condo stock is much more intertwined with post-2000 redevelopment, while the single-family stock is predominantly older and much slower-changing.**

That age difference later lines up closely with turnover, ownership, and market-circulation results.

Figure 7. Building-age maps

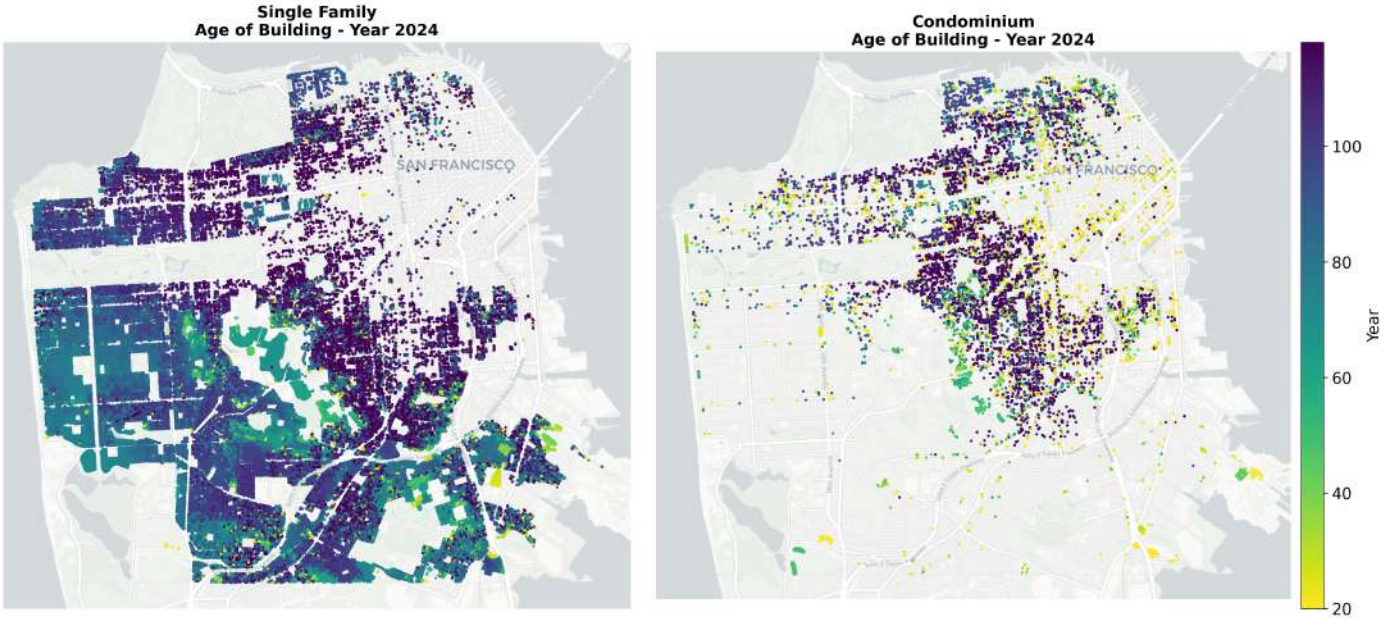


Table 11: Oldest single-family neighborhoods

Neighborhood	Median building age	Property count
6C Lower Pacific Heights	124 years	345
6B Hayes Valley	124 years	221
6F North Panhandle	124 years	144
9C Inner Mission	124 years	1064
5B Haight Ashbury	124 years	133
7B Pacific Heights	118 years	1032
8G Telegraph Hill	117 years	106
5C Noe Valley	116 years	2825
5K Eureka Valley/Dolores Heights	116 years	1498
5F Buena Vista Park/Ashbury Heights	114 years	442

Table 12: Youngest single-family neighborhoods

Neighborhood	Median building age	Property count
10J Hunters Point	36 years	146
4B Diamond Heights	59 years	475
4D Forest Knolls	63 years	475
4K Sherwood Forest	66 years	230
4F Midtown Terrace	67 years	975
5H Clarendon Heights	70 years	288
2A Golden Gate Heights	73 years	1325
3A Lake Shore	74 years	1135
6A Anza Vista	76 years	112
4H Miraloma Park	77 years	1954

Table 13: Oldest condominium neighborhoods

Neighborhood	Median building age	Property count
6E Alamo Square	124 years	292
5J Duboce Triangle	118 years	406
5B Haight Ashbury	117 years	667
7C Presidio Heights	115 years	339
5K Eureka Valley/Dolores Heights	113 years	1470
5E Cole Valley/Parnassus Heights	112 years	632
6F North Panhandle	107 years	1122
1D Lake Street	103 years	523
5C Noe Valley	99 years	1752
7D Cow Hollow	98 years	1056

Table 14: Youngest condominium neighborhoods

Neighborhood	Median building age	Property count
10J Hunters Point	8 years	581
3D Stonestown	9 years	122
9J Central Waterfront/Dogpatch	12 years	715
10A Bayview	14 years	215
9B Financial District South	15 years	2438
9D Mission Bay	15 years	2223
10M Candlestick Point	17 years	375
9H South Beach	18 years	3671
9F South of Market	20 years	1886
8J Tenderloin	21 years	406

4D. Where are the recently sold homes?

Because Proposition 13 anchors many long-held assessments to old base-year values, the most market-relevant subset of the roll is often the housing that has sold recently. This section isolates homes sold in the **last five years** and examines their 2024 assessed values. The goal is not to reconstruct transaction prices, but to narrow attention to properties whose assessments are more likely to be closer to current conditions.

Key findings

Section 4D is the bridge from neighborhood geography to market structure.

- For **single-family homes**, the neighborhoods with the largest number of recent transactions include **Bernal Heights**, **Noe Valley**, and **Excelsior**. Their recently sold medians are much closer to contemporary expectations than the all-stock medians.
- For **condominiums**, recent activity is concentrated in **South Beach**, **Pacific Heights**, **Inner Mission**, **Mission Bay**, and nearby dense urban neighborhoods. This is already a sign that the condo segment is carrying a disproportionate share of current market flow.
- The recent-vs-all scatter makes the Proposition 13 effect visible. Neighborhoods with a large long-held stock can look artificially cheap on the roll because many parcels are still anchored to older assessments.
- That means the assessor roll is strong for **relative comparison** and **spatial structure**, but it must be handled carefully for statements about the current absolute price level.
- Filtering to the last five years gives a clearer view of the **current marginal market**: the homes that are actually resetting assessments and absorbing present-day demand.
- The trade-off is selectivity. This filter improves market relevance but narrows the analysis to a thinner subset of stock.

That final point sets up the next section. Once the report separates **total stock** from **recently circulating stock**, it becomes possible to ask which housing segments actually turn over and what kind of post-purchase occupancy they appear to produce.

Figure 8. Recently sold homes in the last five years

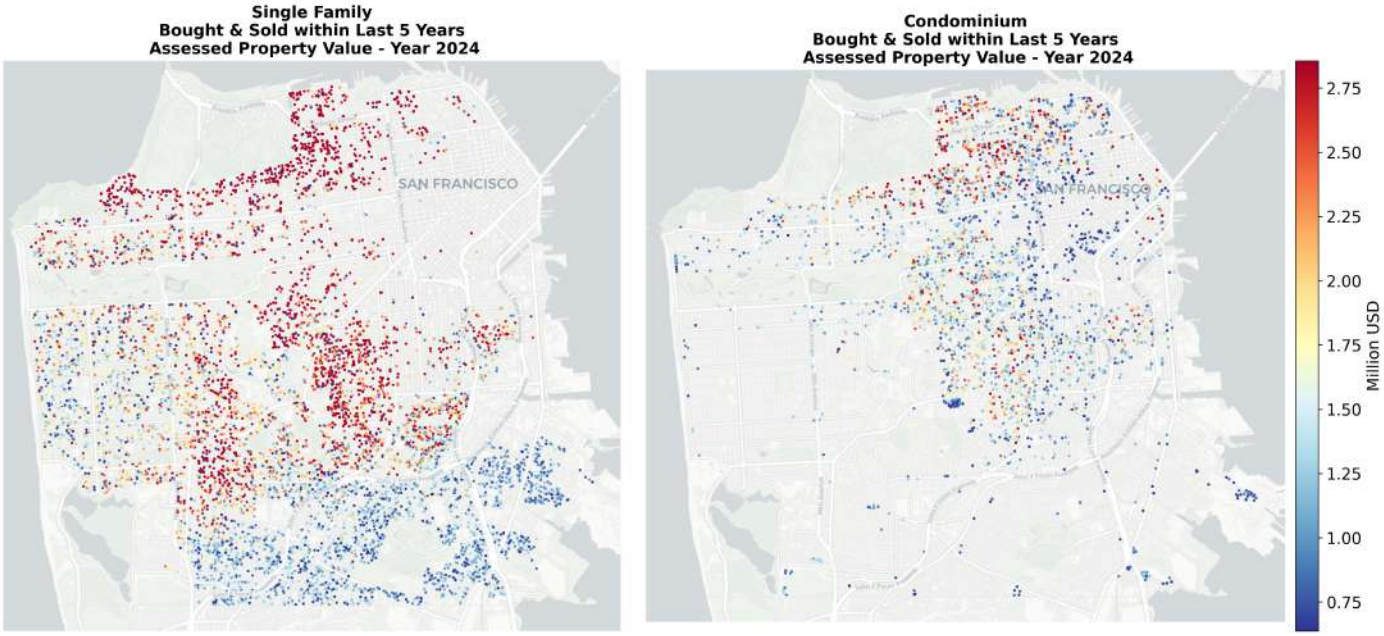


Figure 9. Proposition 13 effect on neighborhood medians

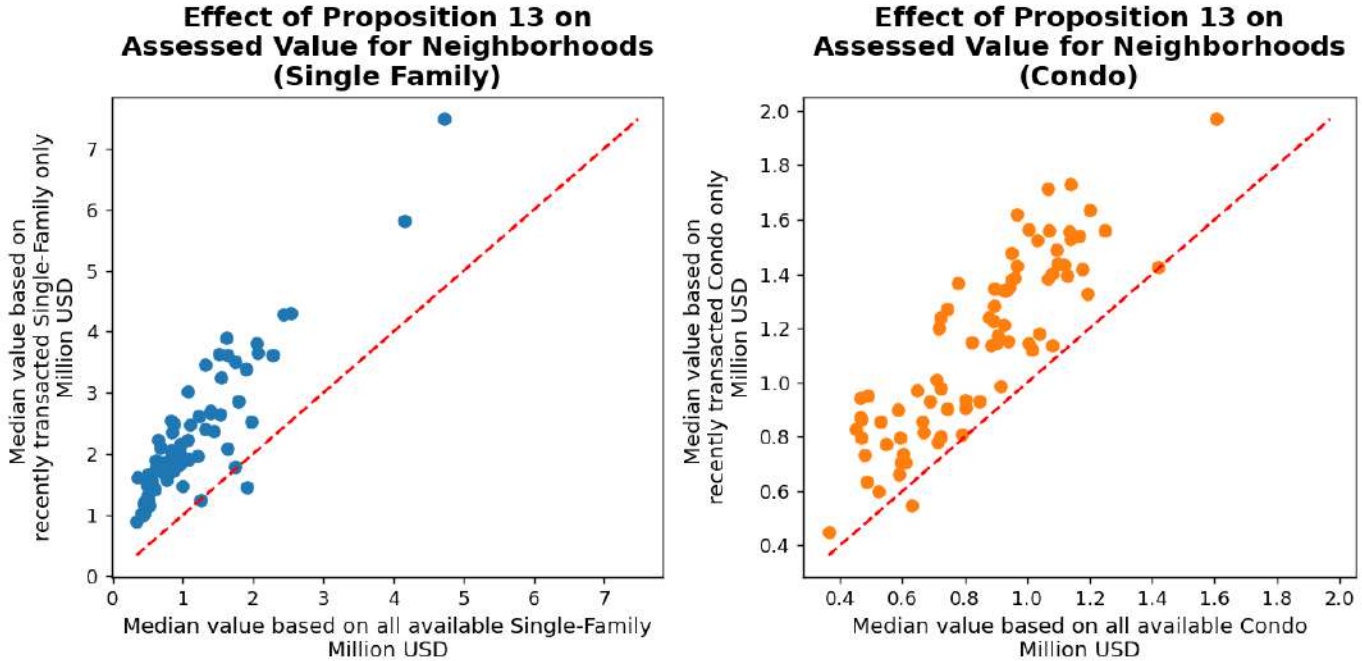


Table 15: Single-family neighborhoods with highest number of recent transactions

Neighborhood	Assessed value (USD millions)	Recent transaction count
9A Bernal Heights	$1.683M	589
5C Noe Valley	$2.679M	412
10C Excelsior	$1.171M	358
2D Parkside	$1.553M	353
2E Central Sunset	$1.665M	318
2B Outer Parkside	$1.486M	305
2C Outer Sunset	$1.559M	275
5A Glen Park	$1.925M	256
10A Bayview	$1.008M	237
10F Portola	$1.244M	236

Table 16: Condominium neighborhoods with highest number of recent transactions

Neighborhood	Assessed value (USD millions)	Recent transaction count
9H South Beach	$1.395M	793
7B Pacific Heights	$1.558M	594
9C Inner Mission	$1.144M	491
8F Van Ness/ Civic Center	$0.815M	457
9D Mission Bay	$1.137M	447
9B Financial District South	$1.419M	428
6B Hayes Valley	$1.139M	398
5C Noe Valley	$1.561M	377
9F South of Market	$0.780M	371
8E Russian Hill	$1.438M	350

4E. Market composition, turnover, and buyer profile

This section brings together the key measures behind the **Two Housing Circuits** interpretation:

- the size of each housing stock,
- the share that appears owner-occupied,
- the number of homes sold in the last five years,
- and the occupancy profile of those recent purchases.

The homeowner exemption should be treated as a **proxy** for owner occupancy, not a perfect legal classification. Even so, once it is combined with turnover, it becomes highly informative.

Key findings

This is the section brings up the report's central argument.

- The analysis set contains **94,844 single-family homes** and **52,042 condominiums**. Single-family housing dominates the stock in San Francisco (**roughly a 2:1 ratio of single-family versus condos**).
- But the flow story is different. Only **8,531 single-family homes sold** in the last five years (**9.0% of stock**), while **10,520 condos sold** (**20.2% of stock**). The smaller condo segment generated more sales than the much larger single-family segment.
- Owner occupancy is **55.6% for single-family homes** and **34.2% for condos**. That gap shows the single-family stock is more rooted in long-duration household holding, while condos are more exposed to rental and investment use.
- Among recent purchases, the contrast narrows sharply: only **27.8%** of recently purchased single-family homes and **24.0%** of recently purchased condos appear owner-occupied. In both segments, most recent purchases appear to sit outside the owner-occupied category after sale.
- In other words, **for every 4 properties transacted in San Francisco, 3 are bought by investors and 1 is bought by onwer-occupier**.
- This is the key structural result: **single-family homes dominate the stock, but condos dominate the flow**.
- It also suggests that **investor or non-owner-occupying demand is not confined to condos**. Once a home enters the active market, both single-family homes and condos appear to face substantial competition from buyers who do not immediately occupy the property.

Read together, these results support the **Two Housing Circuits** interpretation.

San Francisco has **Two Housing Circuits**:

- The **Legacy Circuit** is larger, older, more owner-oriented, and slower to recycle.
- The **Circulation Circuit** is smaller, but it turns over more often and does more of the work of absorbing current demand.

Figure 10. Market composition, turnover, and buyer profile

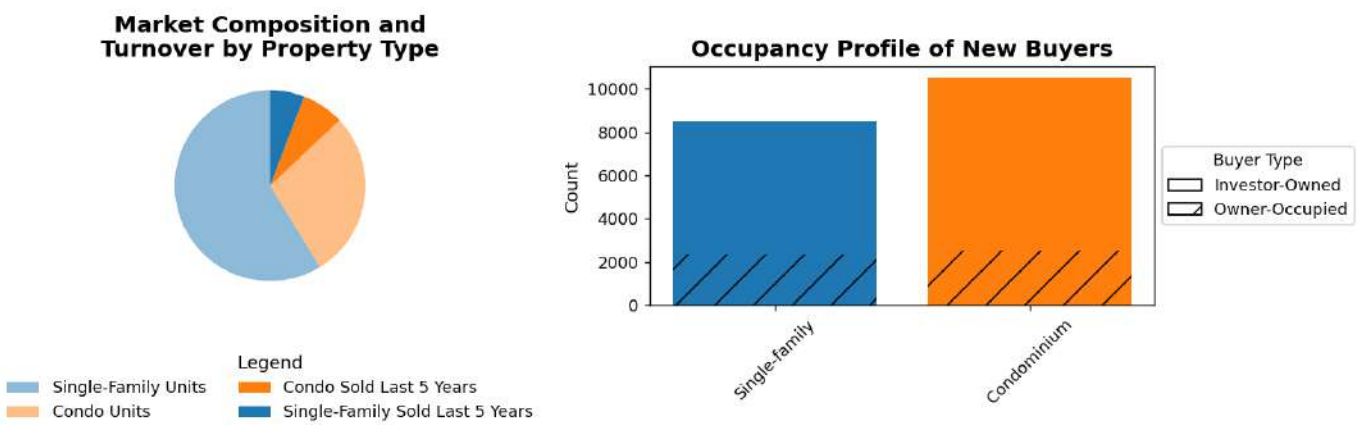


Table 17: Occupancy mix among units purchased in the last 5 years

Housing type	Units sold (last 5 yrs)	Owner-occupied purchases	Owner-occupied share	Renter-occupied purchases	Renter-occupied share
Single-family	8,531	2,370	27.8%	6,161	72.2%
Condominium	10,520	2,520	24.0%	8,000	76.0%

Table 18: Occupancy and recent turnover by housing type

Housing type	Units analyzed	Owner-occupied units	Owner-occupied share	Units sold (last 5 yrs)	Turnover rate (last 5 yrs)
Single-family	94,844	52,692	55.6%	8,531	9.0%
Condominium	52,042	17,800	34.2%	10,520	20.2%

4F. Number of years since last sale

Section 4E showed that the single-family stock is larger but much less active, while the condo stock is smaller but more market-facing. This section checks that interpretation directly through **years since last sale**.

The question is simple: do the two property types show the same ownership rhythm, or do they follow different turnover structures? This is where the distinction between **legacy stock** and **circulating stock** becomes especially visible.

Key findings

Section 4F turns the turnover argument into a dynamic one.

- The **condo distribution** looks much more like a circulation market. Many units sold recently, and the counts thin out in a comparatively regular way as holding periods lengthen.
- The **single-family distribution** does not. It contains a much heavier long-duration component, consistent with a large stock that stays in the same ownership for extended periods.
- The decay fit sharpens the distinction. For **condominiums**, the fitted decay parameter is $\lambda = 0.078$, implying an estimated half-life of about **8.9 years**. That is strong evidence of a genuine circulation process.
- For **single-family homes**, the fitted parameter is much flatter at $\lambda = 0.015$, with a notional half-life of about **46.2 years**. More important than the number itself is the weak fit: the single-family stock does not behave like clean recycling inventory.
- In practical terms, condos behave like a segment that continually re-enters the market, while a large part of the single-family stock behaves like **legacy housing** — present in the city, but only weakly present in the active market.

This is why the distinction between **total housing stock** and **circulating housing stock** matters so much. In a market where prices are set at the margin, the relevant supply is not simply the number of homes that exist. It is the number of homes that actually circulate.

Figure 11. Years since last sale maps

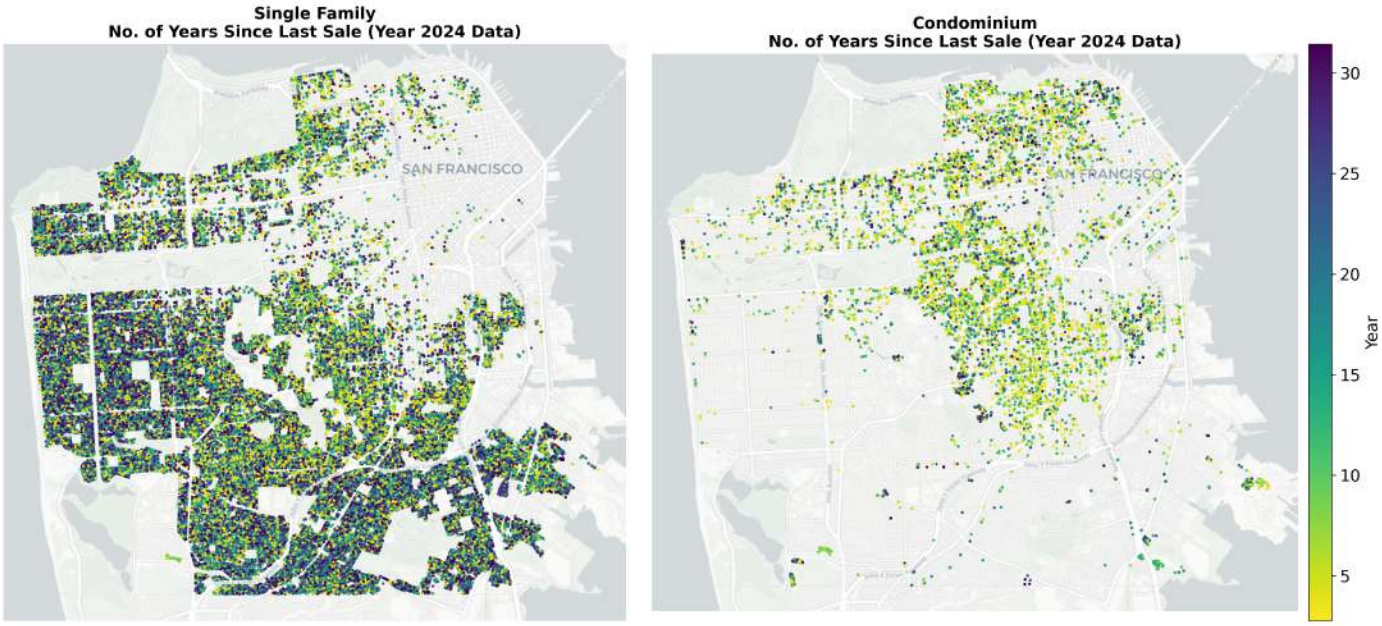


Figure 12. Holding-period histograms

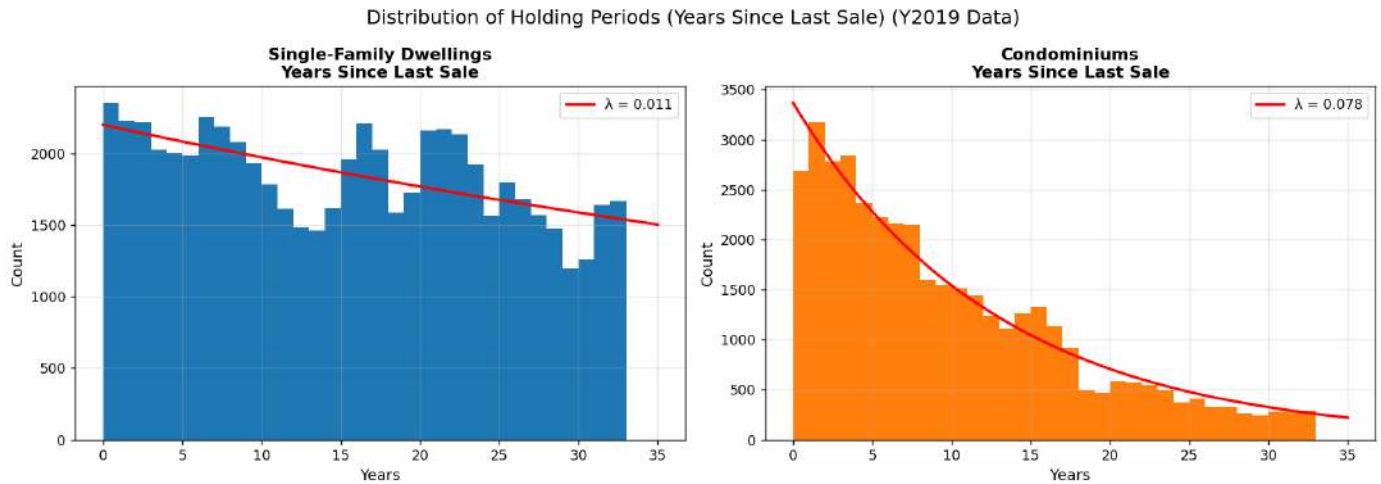


Figure 13. Decay-function fit comparison

Decay-function interpretation

The condo distribution looks much more like a **circulation market**:

- many units sold recently,
- counts fall away steadily as holding periods lengthen,
- and the histogram broadly resembles a time-dependent exponential decay process.

The single-family distribution does **not** behave that way nearly as cleanly.

- It suggests a large stock of homes that remain in the same ownership for very long periods.
- It looks less like churn-driven inventory and more like an embedded stock with weak recycling.

The exponential decay fit captures not only how many homes sold recently, but how quickly ownership cohorts thin out as time passes.

For **condominiums**:

- fitted decay parameter: $\lambda = 0.078$
- intuitive interpretation: the count of condos remaining in a given "not sold since" cohort shrinks by about **7.8% per additional year**
- implied half-life: about **8.9 years**

This is strong evidence of a real **Circulation Circuit**. Condos behave like a housing segment that continually recycles through the market.

For **single-family homes**:

- fitted parameter: $\lambda = 0.011$
- implied notional half-life: about **63.0 years**
- but the fit is weak and should be interpreted cautiously

This shows that a substantial share of the single-family stock is not behaving like ordinary circulating inventory at all. Instead, much of it behaves like **legacy housing**: long-held owner-occupied residences, or long-held rental assets retained over decades.

Conclusions from Sections 4E and 4F

A large portion of San Francisco's single-family stock is effectively not for sale in any normal market sense.

The real constraint is not simply how many homes exist. It is how many homes are actually circulating.

In a city where prices are set at the margin, this matters enormously. The circulation stock — rather than the full physical stock — is where price discovery, entry competition, and investor pressure are concentrated.

Broader civic interpretation

- The **Legacy Circuit** preserves rootedness, continuity, and neighborhood memory.
- The **Circulation Circuit** enables entry, adaptation, and demographic renewal.

A vibrant city needs both. But if too much of the stock remains locked in the legacy circuit, the city becomes increasingly hard for newcomers to enter. If the circulation circuit becomes too investor-oriented, access remains available mainly as a rental or investment channel rather than as long-term ownership opportunity.

Overall Framework

Taken together, the notebook is best understood through the framework of **Two Housing Circuits**:

The Legacy Circuit

This consists of homes that are:

- older,
- tightly held,
- often owner-occupied or long-term retained,
- slow to re-enter the market,
- and structurally important to neighborhood continuity.

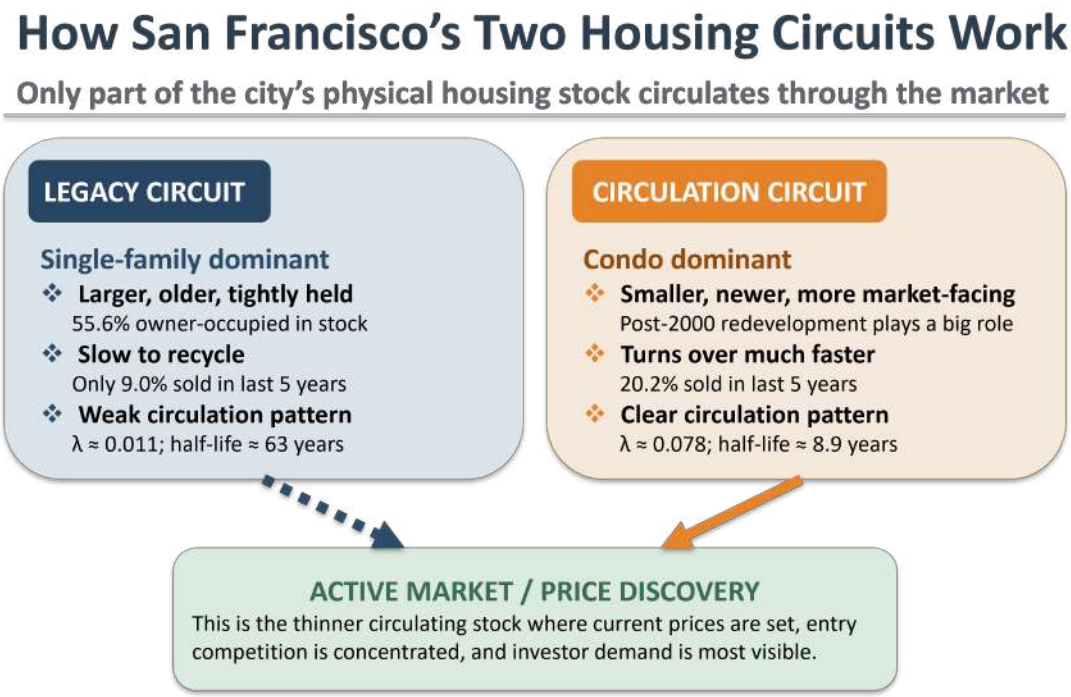
The Circulation Circuit

This consists of homes that:

- turn over more frequently,
- absorb new demand,
- play a disproportionate role in price discovery,
- and determine who can still enter the city as a homeowner.

This framework connects the report's maps, neighborhood rankings, age patterns, turnover summaries, and decay fits into one coherent structural interpretation.

Figure 14. San Francisco's Two housing Circuits



Final synthesis of 7 major conclusions

1. San Francisco's housing geography is sharply unequal

The city contains large neighborhood gaps even within the same property type. Single-family medians range from **$4.729M in Presidio Heights** to **$0.342M in Hunters Point**. On a per-square-foot basis, single-family values range from about **$1,199/sq ft in Telegraph Hill** to **$253/sq ft in Hunters Point**. These are distinct submarkets, not small deviations around a citywide average.

2. Single-family homes and condos play different structural roles

Across the report, the same contrast appears repeatedly.

- **Age:** single-family neighborhoods are often much older; many condo-heavy districts are products of more recent redevelopment.
- **Turnover:** condos show a **20.2%** five-year turnover rate versus **9.0%** for single-family homes.
- **Occupancy:** apparent owner occupancy is **55.6%** for single-family homes and **34.2%** for condos.
- **Recent purchases:** only **27.8%** of recently purchased single-family homes and **24.0%** of recently purchased condos appear owner-occupied.
- **Holding periods:** condos show a much clearer circulation pattern with $\lambda = 0.078$ and an implied half-life of **8.9 years**, while single-family homes show a much flatter pattern with $\lambda = 0.011$ and a notional half-life of **63.0 years**.

These results support the report's core interpretation: **single-family homes behave more like the legacy circuit, while condos behave more like the Circulation Circuit**.

3. The active market is much smaller than the physical stock

This is the most important practical result in the report.

San Francisco contains about **94,844** single-family homes, yet only **8,531** sold in the last five years. By contrast, the much smaller condo stock of **52,042** units generated **10,520** sales over the same period. The homes that actually absorb demand and help set current prices come from a much thinner pool than headline stock counts suggest.

4. Proposition 13 is not a background detail; it shapes the reading of the entire dataset

The recent-sales comparison shows that all-stock neighborhood medians can materially understate current conditions in neighborhoods with many long-held properties. That does not make the dataset less useful. It means the roll should be read as both a valuation record and a record of ownership history.

5. Investor-oriented outcomes appear in both property types

A striking result is that recently purchased units in **both** segments appear overwhelmingly outside the owner-occupied category. The implication is that investor or non-owner-occupying demand is not confined to condos. Once a home enters the active market, both property types appear exposed to similar competitive pressure.

6. The main bottleneck is circulation, not just stock

The housing problem described here is not only about how many units exist. It is also about how many units actually return to market. The condo segment does far more of that work. The single-family segment contains a much larger share of long-held stock that contributes relatively little to current market access.

7. The Two Housing Circuits framework ties the report together

The value maps, age patterns, recent-sales filter, turnover tables, buyer-profile proxy, and holding-period distributions all point in the same direction.

- The **legacy circuit** is older, more tightly held, and slower to recycle.
- The **Circulation Circuit** turns over more often, carries more of the burden of entry, and is where competition is most concentrated.

That is the main conclusion of the project. San Francisco's housing challenge is not only that housing is expensive. It is that a large share of the stock is structurally slow-moving, while the smaller circulating share bears most of the pressure of entry, investment demand, and price discovery.